Analyzing Dose Reduction in PET-MRI: Insights from State-of-the-Art AI Restoration Techniques

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The findings from Yan-Ran Wang et al. (2023) [1], a pioneering study comparing state-of-the-art AI models for restoring low-count PET-MRI images across the entire dose reduction spectrum, are analyzed. These models enable radiotracer dose reductions while preserving diagnostic image quality (DIQ) for improved patient safety. Results show that 6.25% dose reductions are achievable, with SwinIR delivering the highest structural similarity index (SSIM), peak signal-to-noise-ratio (PSNR) and VIF (visual information fidelity), while U-Net offers comparable DIQ. Despite these advancements, challenges remain, including ensuring model generalizability with diverse datasets and and developing tailored loss functions for diagnostic radiology.

Keywords: PET-MRI, low-count imaging, CNN, Transformer.

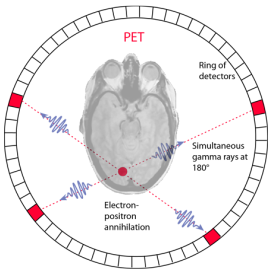
1. Introduction

The combination of positron emission tomography (PET) and magnetic resonance imaging (MRI) into PET-MRI provides higher imaging capabilities with reduced radiation exposure compared to PET-CT. However, further lowering the radiotracer dosage in PET-MRI decreases the diagnostic image quality (DIQ) due to the increase in noise and image contrast. To address this challenge, state-of-the-art AI models have been developed. Yan-Ran Wang et al. (2023) [1] were the first to provide a comprehensive comparison of such models using quantitative and qualitative metrics across the whole dose spectrum reduction of the standard dose. This study will be analyzed and further discussed herein.

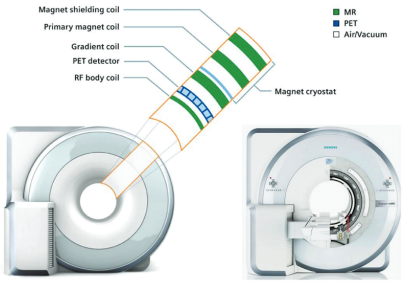
1.1. PET-MRI Technology Overview

Positron Emission Tomography (PET) is an imaging technique that uses *radiopharmaceuticals* [1] as tracers to visualize and measure metabolic activities within the body. When the radiotracer undergoes beta plus decay, it emits a positron which in turn reacts with an electron and generates two gamma rays in opposite directions [2]. Finally, the gamma rays are detected by a ring of sensors (Figure 1a) and the information is sufficient to form a three-dimensional image with the concentration of the radioisotope in each location. Its integration with x-ray computed tomography (CT) into PET-CT enables simultaneous imaging of metabolic and anatomical data by superposing their images [3].

On the other hand, magnetic resonance imaging (MRI) is a medical imaging technique that uses strong magnetic fields, magnetic field gradients and radio waves to form the anatomical images of the body. Differently from CT and PET, MRI does not use x-rays and ionizing radiation and, consequently, has little biologically harmful effects [4]. Inspired by the PET-CT, the combination of PET and MRI into PET-MRI (Figure 1b) became available nearly a decade after [5], overcoming challenges like developing a PET sensors compatible with strong magnetic fields and a magnetic resonance scanner with a stable magnetic field in the presence of a PET insert. This integration not only reduces radiation exposure but also enhances soft tissue contrast, making it ideal for *oncology* [2], neurology, and cardiology [6].



(a) Image from HyperPhysics illustrating how PET works [2].



(b) Diagram of a PET-MRI scanner. Image courtesy of Siemens Healthcare, Erlangen, Germany.

Figure 1

1.2. Advantages of Enhancing Low-Count PET Imaging

One of the most common used radiopharmaceuticals is the F-FDG (Fluorodeoxyglucose labeled with Fluorine-18). Since glucose is actively absorbed by metabolically active cells, F-FDG accumulates in tumor cells, making it highly effective for detecting cancerous tissue [7]. However, this procedure emits ionizing radiation, posing risks to pa-

[1] Drugs containing radioactive isotopes.
[2] A branch of medicine that specializes in the diagnosis and treatment of cancer.

tients, particularly for repeat imaging and pediatric patients [1]. Even low-level exposure may lead to stochastic effects, such as cancer, often occurring several years to decades later [8]. PET-MRI addresses this critical issue in medical imaging by eliminating the the CT component in PET-CT, responsible for up to 70% of total radiation exposure [9]. Nevertheless, even low radiation exposure can increase cancer risks (proportionally to the *absorbed dose* [3] in the tissue [8]). Therefore, lowering the dose of the radiotracer while maintaining image quality is a critical challenge in PET-MRI.

Furthermore, low-count PET can considerably cut costs by decreasing the amount of F-FGD required. The cost are mainly influenced by the radiopharmaceuticals, which average $528 per dose (38.9% of total scan costs) and range from $150 to $922 [10]. For instance, reducing the dose to 6.25% of the standard could lower total scan costs by 17%, considering 46% of F-FDG production costs are material-related[4].

Finally, techniques for restoring low-count PET images are also applicable to short-duration PET scans, which face similar challenges due to fewer data statistics. These would enable an increase in patient throughput without compromising image quality.

1.3. Challenges in Low-Count PET Imaging

Reducing the radiotracer dosage decreases the number of positron annihilation events, directly impacting the diagnostic image quality (DIQ). This leads to increased image noise, which obscures important details due to pronounced random variations; a lower *contrast-to-noise ratio (CNR)*[5], diminishing the differentiation between regions of interest (ROIs), such as tumors and surrounding tissues; and bias in uptake measurements, which quantify metabolic activity [12].

Simple post-processing operations, such as denoising filters, are insufficient due to their reliance on smoothing or averaging neighboring pixel intensities. Hence, they fail to recover missing or biased information caused by the lack of initial data. In contrast, learning-based reconstruction methods offer a promising approach by learning patterns from large datasets and can construct relations between low- and full-count images. This enables them to enhance the images to identify not only noise but also structural inconsistencies within it [12]. Among these methods, AI-powered deep learning techniques are particularly effective.

2. AI Models for PET-MRI Image Restoration

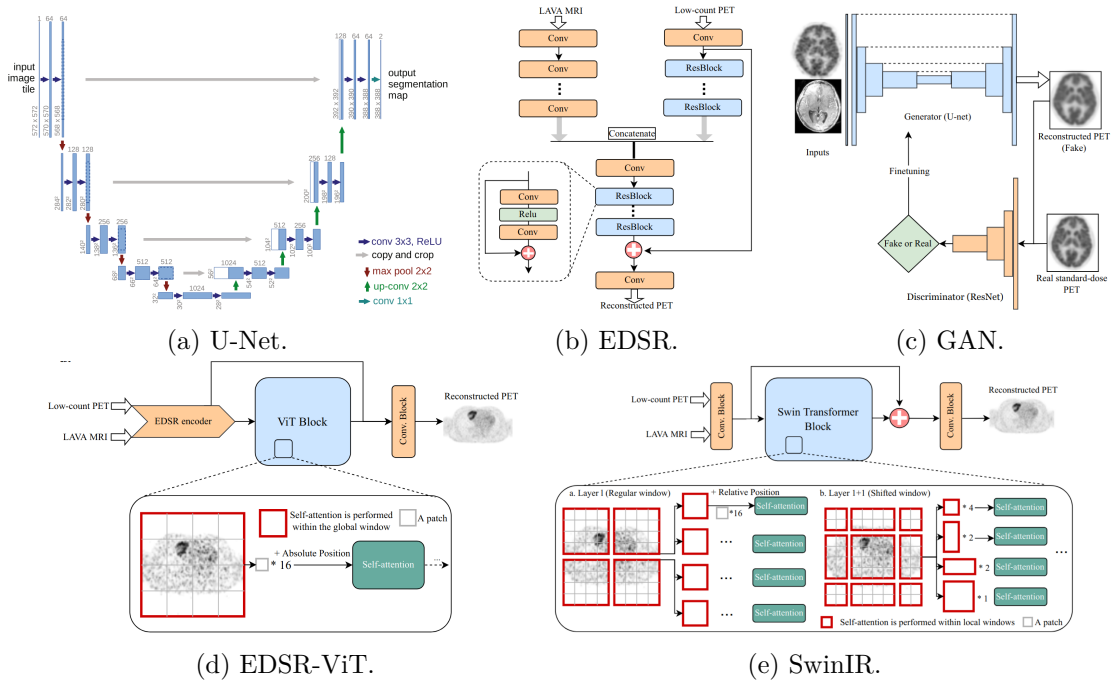
Many AI models were built to reconstruct low-count PET scans and five of the most advanced models (Figure 2) were analyzed in the recent study for their performance at different low doses [1]. There were three convolutional neural networks (CNNs): U-Net [13], enhanced deep super-resolution network (EDSR) [14] and generative adversarial network (GAN) [15]; and two transformer models: EDSR-ViT [1] and SwinIR [16].

[3] Defined as the radiation energy absorbed in a small volume element of matter divided by its mass.

[4] A rough estimate that considers a reduction in the production costs for fluoride-18 and the synthesis of FDG. Such vales were obtained as a percentage of the total costs of various centers across Europe. [11]

[5] Quantifies the contrast between features relative to noise unlike SNR, which measures signal strength.

The key difference between CNNs and transformer models is that CNNs use layers of filters to capture local spacial features while transformer models, originally developed for natural language processing (NLP), use self-attention mechanisms to capture global dependencies across the entire input.



(a) U-Net. (b) EDSR. (c) GAN.

(d) EDSR-ViT. (e) SwinIR.

Figure 2: AI model architectures to reconstruct low-count images. Reprinted, with permission, from a) Olaf Ronneberger et al. [13], and b-e) Yan-Ran Wang et al. [1].

2.1. U-Net

The architecture of the U-Net (Figure 2a) consists of a contracting path (encoder) on the left and an expansive path (decoder) on the right. Each step of the contracting path is a repeated application of two 3x3 convolutions with a *rectified linear unit (ReLU)* [6] and a 2x2 *max pooling operation* [7] with stride two for downsampling. Contrastingly, in the expansive path, every step consists of an upsampling followed by a 2x2 convolution ("up-convolution"), a concatenation with the symmetrical feature map in the left, which is essential for linking the expansive path semantic information to the contracting path spatial information, and two 3x3 convolutions also with ReLU. Finally, a 1x1 convolution is used at the top to get the desired output [13].

[6] Simple activation function that works as a ramp and is widely used in deep learning for introducing non-linearity to the model, what enables it to solve more complex problems [17].

[7] A network layer that downsamples and aggregates information, reducing the amount of computation and memory required in the following steps [18]. In this case, the *max* operator was used.

2.2. EDSR

The EDSR is a residual network, which is an architecture that introduces *shortcut connections* to bypass one or more layers, allowing the network to learn residuals and solve the *vanishing gradient problem* [8] during training [20]. It was based on ResNet and SR-ResNet and optimized by removing unnecessary modules to simplify its architecture. Its basic block contains only two convolutions, a ReLU and an add residual. Hence, EDSR removed batch normalization and the follow-up ReLU block. Moreover, it was further adapted for the low-count PET scan study as seen in Figure 2b.

2.3. GAN

GAN is an architecture (Figure 2c) that consists of two models: a generative and a discriminative. The first one is responsible for generating synthetic instances of data, while the latter for estimating the probability that the sample is a training data instead of a synthetic one. They are trained simultaneously with the objective of the generative model fooling the discriminative into making a wrong estimation [15].

The generative architecture has five strided convolutional transpose layers ("deconv") with 2d batch norm and ReLU activation. These transpose layers expands the data to create the image. In contrast, the discriminator architecture has five strided convolution layers with batch normalization and Leaky ReLU, such that each layer reduces the spacial dimensionality of the data. The last layer has a Sigmoid function, which is responsible for generating the likelihood probability of the image being real or synthetic [15].

2.4. EDSR-ViT

Vision transformer (ViT) is a type of transformer adapted for image processing and it works by splitting the image into fixed-size patches, converting each patch into a vector, adding positional embeddings and inputting the sequence into a Transformer encoder [21]. One important feature of this architecture is that ViTs can capture long-range dependencies in images, unlike the other CNNs previously mentioned. Moreover, it was first used in image restoration in EDSR-ViT, which attaches the adapted EDSR CNN encoder previously described to the transformer block (Figure 2d) [1].

2.5. SwinIR

The SwinIR is a model specifically designed for image restoration that integrates the advantages the Swin Transformer and a CNN. The Swin Transformer works by dividing the images into non-overlapping windows. Within each, self-attention is computed to capture local dependencies. Furthermore, the windows are shifted to subsequent layers to create overlaps and enable cross-window connections. This mechanism allows the model to balance computational efficiency and the ability to capture global dependencies in the images [22], which were specially important to improve the contrast of structural details.

[8] This issue arises during backpropagation and worsens with the increase in depth. Weight updates rely on the gradient value, which can diminish considerably in deeper networks, hindering learning [19].

It consists of three modules (Figure 2e): shallow feature extraction, deep feature extraction and high-quality (HQ) image reconstruction. The shallow feature extraction uses a single convolutional layer to capture low-level features. The deep feature extraction has multiple Residual Swin Transformer blocks (RSTBs) with self-attention mechanism to model local and global dependencies. Finally, the HQ image reconstruction has a convolution layer to synthesize the output image aggregating the shallow and deep features [16].

3. Comparative Analysis of AI Models

Yan-Ran Wang et al. (2023) [1] provided a comparison of the state-of-the-art AIs for reconstructing low-count PET-MRI images while maintaining high diagnostic image quality (DIQ). To achieve this, the authors created a comprehensive evaluation framework and applied it to test each AI model across the whole dose reduction spectrum (75%, 50%, 25%, 12.5%, 6.25% and 1% of the clinical 3 MBq/kg ^{18}F-FDG dose).

The study was based on scans of 32 children and young adults lymphoma patients from two different cohorts. Each scan was used to simulate low-count PET images instead of injecting multiple radiotracer doses in a single patient. Totaling 476 PET-MRI scans (336 from the primary Stanford cohort and 140 from the Tübingen external cohort). Each AI architecture presented was trained and tested separately in the primary Stanford cohort for each dose reduction percentage (resulting in 30 AI models). Furthermore, they were later tested in the Tübingen external cohort to generalize the findings.

3.1. Evaluation Framework

The evaluation framework consisted of three quantitative metrics to measure the quality of the restored low-count PET images: SSIM (the structure similarity index), PSNR (peak signal-to-noise-ratio) and VIF (visual information fidelity). SSIM was used to measure the similarity between the AI-restored to the original standard dose PET images, considering a combination of structure, luminance, and contrast, with a scale ranging from 0 (no similarity) to 1 (perfect similarity). This metric emphasizes structural information by measuring dependencies between pixels in close spatial proximity. Similarly, PSNR measures the ratio between the maximum possible power and the noise power affecting the image, where a higher PSNR indicates better reconstruction of the image to match the original one. In contrast, VIF uses natural statistics models to mimic the way humans process visual data, emphasizing structural and perceptual features rather than pixel-wise similarity. It also ranges from 0 (no similarity) to 1 (perfect similarity).

Moreover, two board-certified radiologists reviewed the standard dose, low-count and AI-restored images by SwinIR and U-Net (were chosen to represent the transformer and CNN families, respectively) to assess their DIQ, using a 5-point Likert scale. Finally, for quantitative measures of tumor metabolism, standardized uptake values (SUVs) were used, a widely accepted metric for measuring radiotracer concentration in specific ROIs. In this study, SUVs were measured in tumors using the liver as an internal reference

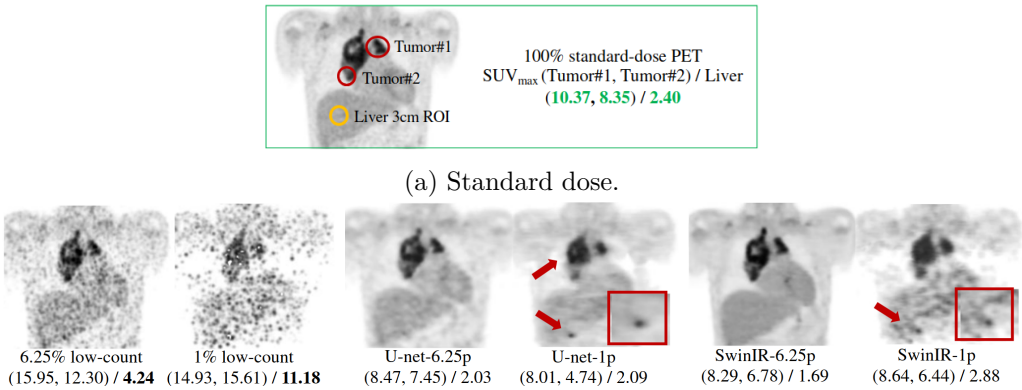
standard. The proximity of the SUV in the low-count AI-reconstructed image to that of the full-count image indicates the accuracy of the method.

3.2. Results

The mean values of the performance metrics SSIM, PSNR and VIF were calculated for both cohorts for the whole dose reduction spectrum. The improvements were more significant at the 6.25% dose and were further detailed in Table 1. Between the five AI models used in the study, the SwinIR achieved the highest SSIM, PSNR and VIF on the primary cohort and the highest SSIM and VIF in the secondary cohort, which demonstrates the generalization between the cross-continental cohorts.

The quantitative improvements in the 75% and 50% low-count PET images were not significant, probably because of their similarity to the original full-count scan. Moreover, there was a steep drop in SSIM, PSNR and VIF from 6.25 to 1% doses.

Low-count PET images showed higher noise and information loss, increasing the SUV_{max} values in the liver and tumors (Figure 3b). The AI models effectively reduced these discrepancies, bringing the values closer to those observed in the full-count value image (Figures 3c and 3d). However, extremely ultra-low-count 1% restored images also demonstrated an enlargement of false upstaging in the liver area. Such effects could impact clinical decisions with false-negative and false-positives and highlights the need for cautions use of AI to balance improved image quality and diagnostic risks.



(a) Standard dose.

(b) Original low-count. (c) U-Net restored. (d) SwinIR restored.

Figure 3: SUV_{max} comparison for two tumors and the liver. Detailed results covering the entire dose reduction spectrum are presented in Figure 4.

At last, the radiologists' assessment (Table 2) showed that both SwinIR and U-Net can significantly improve diagnostic quality of low-count PET images. Also, across the dose reduction spectrum (except at 1%) SwinIR achieved 1/3 false-negative/false-positive rates and U-Net had 0/3, while the original low-count images recorded 12/24. Notably, at 6.25%, both models performed impressively well, with SwinIR achieving 0/1 and U-Net 0/0, compared to the original low-count images, which had 6/10.

4. Challenges for AI in low-count PET Reconstruction

Despite the advancements demonstrated by the models in low-count PET reconstruction, some challenges remain that limit their effectiveness. For instance, pre-trained models, such as GAN and EDSR-ViT, rely on specific datasets that might limit their generalizability to different radiotracers, patients and anatomical parts. To overcome this issue, future models should incorporate diverse multi-institutional data and remove *compounding factors* [9] to mitigate bias across datasets [23].

In the study, simulated low-count PET images were generated from undersampling of standard dose images for ethically feasibility. Although these simulated data have similar characteristics to the respective low-count images [24], they might not capture subtle clinical details and can bias the training. Hence, further investigation of the AI models' efficiency for real low-count images is needed. Moreover, it only considered scans made with the FDG radiotracer and the results might not be generalizable for non-FDG tracers. Another limitation present in this study is that the mean square loss (MSE loss) and the SSIM loss were were not designed for diagnostic radiology images, but were used in the training of the AI models. This might explain the discrepancies in the restoration quality between different anatomical regions. Consequently, it is necessary to develop specialized *loss functions* [10] for diagnostic radiology images.

Finally, while SwinIR had a superior performance and showed the best quantitative results, it was the most demanding AI model used in the study. With more than 500 Gflops of operations, its training time (129 minutes/epoch) and inference time (122 seconds/subject) were an order of magnitude greater than that of the other models [1].

5. Conclusion

AI models have significantly advanced the restoration of low-dose PET-MRI images. They enable the reduction of the radiotracer dosage while preserving diagnostic image quality. Among the state-of-the-art models analyzed by Yan-Ran Wang et al. (2023) [1], SwinIR consistently outperformed the others in quantitative metrics, such as SSIM, PSNR and VIF across the dose spectrum (Table 1). Additionally, it did not require any pre-training to do the restorations. However, while SwinIR achieved higher quantitative scores, its diagnostic performance was comparable to that of U-Net, which is highly stable and considerably faster. Moreover, the study demonstrated that such AIs are capable of restoring diagnostic image quality at reduced doses up to 6.25%, but achieving viable results at 1% remains beyond the capabilities of current technology. Nonetheless, challenges such as the generalizability of AI models, reliance on simulated data and unsuitable loss functions emphasize the need for further refinement. Developing more robust datasets, incorporating real low-count images and diagnostic radiology-specific tailored loss functions will be key to improving their efficiency.

[9] Dataset variables that may disproportionately influence outcomes (e.g., imaging equipment, patient demographics).

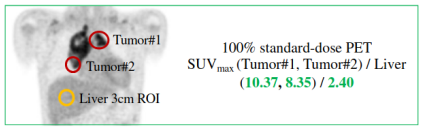
[10] Mathematical functions that measure the difference between the predicted output and the target value, guiding the training process.

A. Supplementary Figures and Tables

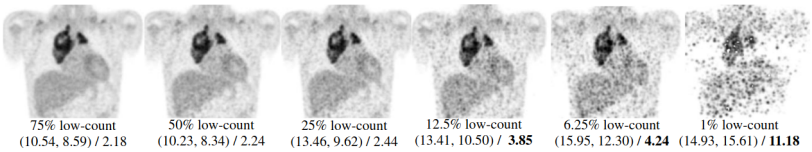
This appendix provides a summary of the results, including images and data, from the AI models' comparison in low-count PET restoration. All content presented here is derived from the study by Yan-Ran Wang et al. (2023) [1], where the complete results can be found.

Table 1: Mean (SD) performance metrics of the five AI models for 6.25% low-count PET reconstruction. Results are shown for the Stanford primary cohort (*test*) and the Tübingen external validation cohort (*external*). Bold values highlight the best performance in each column.

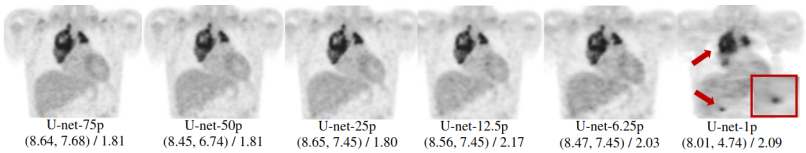
Model	$SSIM_{test}$	$PSNR_{test}$	VIF_{test}	$SSIM_{external}$	$PSNR_{external}$	$VIF_{external}$
EDSR	0.898 (0.033)	39.7 (2.37)	0.454 (0.048)	0.949 (0.020)	39.5 (1.14)	0.466 (0.041)
EDSR-ViT	0.893 (0.035)	38.7 (1.83)	0.433 (0.051)	0.947 (0.021)	38.4 (1.00)	0.436 (0.042)
GAN	0.873 (0.040)	37.4 (2.14)	0.417 (0.047)	0.939 (0.022)	35.7 (0.957)	0.427 (0.039)
U-Net	0.885 (0.036)	39.1 (2.39)	0.442 (0.048)	0.947 (0.020)	**39.6** (1.29)	0.454 (0.042)
SwinIR	**0.910** (0.029)	**39.9** (2.26)	**0.485** (0.046)	**0.950** (0.019)	39.1 (1.08)	**0.483** (0.043)
6.25% low-count PET	0.786 (0.047)	35.0 (2.42)	0.263 (0.046)	0.735 (0.030)	34.9 (1.43)	0.257 (0.030)



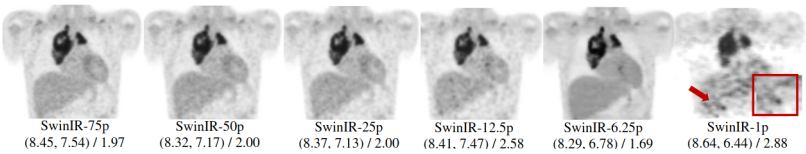
(a) Standard dose PET.



(b) Original low-count PET.



(c) Low-count PET image reconstructed with the U-Net model.



(d) Low-count PET image reconstructed with the SwinIR model.

Figure 4: SUV_{max} comparison for two tumors and the liver across the dose reduction spectrum. Red arrows point to corrupted reconstructions and the red rectangles contains their respective enlargement.

Table 2: Diagnostic image quality (DIQ) for standard-dose, low-count, and AI-restored PET scans.

PET Modality	DIQ Mean (SD)
100% standard-dose PET	5 (0)
75% low-count PET	4.25 (0.463)
SwinIR	5 (0)
U-Net	5 (0)
50% low-count PET	3 (0)
SwinIR	4.5 (0.535)
U-Net	4.25 (0.463)
25% low-count PET	2.625 (0.516)
SwinIR	3.75 (0.463)
U-Net	3.75 (0.463)
12.5% low-count PET	2 (0)
SwinIR	3.25 (0.463)
U-Net	3.625 (0.744)
6.25% low-count PET	1.125 (0.354)
SwinIR	4 (0.926)
U-Net	3.75 (0.886)
1% low-count PET	1 (0)
SwinIR	2.5 (0.534)
U-Net	2.5 (0.534)

DIQ: Diagnostic image quality (1=nondiagnostic, 2=poor, 3=acceptable, 4=good, 5=excellent image quality)

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